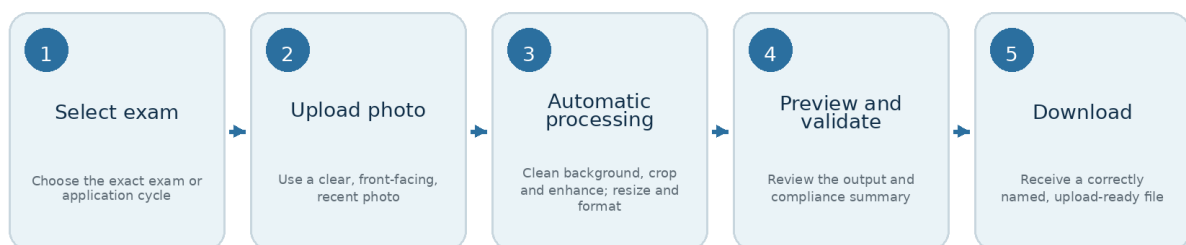


Exam Photo Compliance Platform

A structured foundation for an exam-specific, one-click passport-photo preparation service for Indian application portals.

SELECT EXAM -> UPLOAD PHOTO -> RECEIVE EXAM-READY FILE

The one-minute user journey



Core product promise

The platform converts a decent user photo into a clean, correctly cropped, correctly sized, correctly formatted and correctly named photograph prepared according to the stored requirements of the selected examination.

Document version: 1.2

Status: Foundational product definition

Prepared from the founder's concept brief and subsequent clarifications

How to use this document

This is the starting-point document for product design, engineering, data research, user experience, quality assurance and future commercial planning. It is intentionally more precise than an idea note, but it does not force premature choices about frameworks, vendors or pricing.

Decision language used throughout

CONFIRMED means the direction comes directly from the product concept. RECOMMENDED means a strong implementation proposal that may be changed after prototyping. FUTURE means explicitly outside the first release.

Contents

1. Executive definition and product vision
2. User problem, audience and value proposition
3. Foundational product principles
4. End-to-end user experience
5. Image-processing engine and crop logic
6. Exam-guideline data foundation
7. Validation, quality and failure handling
8. Product scope, MVP and roadmap
9. Recommended system architecture
10. Privacy, security and responsible image processing
11. UX structure and screen requirements
12. Development plan, QA and success metrics
13. Risks, open decisions and decision ledger
14. Appendices: data schema, acceptance checklist and glossary

Important wording

The service can state that an output has been prepared to match stored exam requirements. It should not promise acceptance by an examination authority or third-party portal because the final decision is outside the platform's control.

1. Executive definition and product vision

1.1 Product in one sentence

A web platform where a user selects an Indian examination, uploads a suitable personal photograph, and instantly receives a processed file prepared for that examination's photo-upload requirements.

1.2 The product is not a generic passport-photo tool

The central differentiation is exam awareness. A normal photo editor may remove a background, resize an image or compress a file. This product combines those operations with a structured database of exam-specific requirements and runs them in the correct order as one guided experience.

Generic photo utility	This product
Asks the user to know dimensions and limits	The user selects an exam; the system retrieves the rules
Runs one isolated edit	Runs a complete, ordered compliance pipeline
May compress too much or exceed limits	Targets a high-quality file close to, but below, the permitted ceiling
Leaves naming and format decisions to the user	Exports the required format and filename pattern
Provides an edited image	Provides an exam-contextualized output plus a validation summary

1.3 Vision

Make exam-photo preparation so simple that a candidate does not need to understand pixels, aspect ratios, file-size limits, background cleanup, compression quality or naming conventions. The candidate should only need to know which examination they are applying for and provide a decent source photo.

1.4 Core outcome

The output should look natural, official and recognizably identical to the user while satisfying the stored technical constraints of the selected exam. The experience should feel immediate, predictable and trustworthy.

2. User problem, audience and value proposition

2.1 The user problem

Indian examination application portals frequently specify photo rules using combinations of dimensions, pixel ranges, file-size ceilings, background expectations, formats and naming instructions. Candidates often use several disconnected tools, repeatedly upload files, or sacrifice image quality to make a file small enough. The process is disproportionately confusing for a task that should take less than a minute.

2.2 Primary users

User group	Need	Typical difficulty
First-time applicants	A guided, mistake-resistant process	They may not understand dimensions, KB limits or aspect ratio
Mobile-first candidates	A complete workflow from a phone	Source images may have cluttered backgrounds, shadows or portrait framing
Applicants submitting multiple forms	Fast repeat processing for different exams	Each portal may have different output rules
Assisted-service operators	Consistent output for many candidates	Manual editing is slow and quality varies by operator

2.3 Jobs to be done

- When I am filling an exam form, prepare my photo without making me interpret the technical instructions.
- When my source photo has a normal background, replace it cleanly with the required background.
- When the portal has a strict file-size limit, preserve as much quality as possible without crossing it.
- When the portal specifies a name or extension, give me the exact file I should upload.
- When my original photo is unsuitable, explain the problem clearly and tell me what kind of photo to upload instead.

2.4 Value proposition

For candidates

One exam selection and one upload replace a chain of technical photo-editing tasks.

For the product

The durable asset is the combination of a reliable processing engine and a maintained, versioned exam-guideline database.

3. Foundational product principles

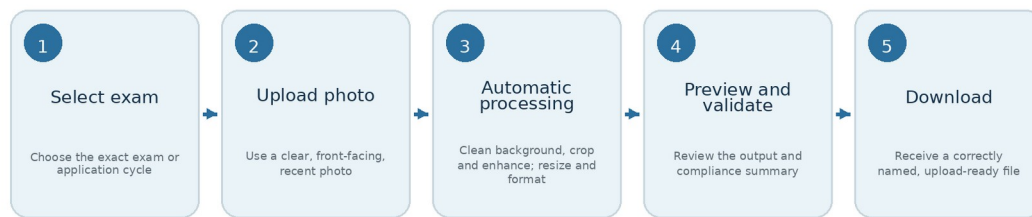
Principle	Product meaning
Exam first, tool second	The user should begin by selecting an exam. The exam record configures the processing pipeline.
Natural crop creates the coverage	Correct framing should naturally produce approximately 75-80% face coverage. The percentage is a universal composition objective, not an overly rigid rejection threshold.
Complete head visibility	Hair at the top and sides, both ears where naturally visible, chin and final beard line must remain visible with a small, natural margin.
Identity must remain unchanged	Enhancement may improve exposure, color, noise and clarity. It must not reshape facial features, slim the face, beautify skin aggressively or generate missing facial detail.
Quality near the permitted ceiling	Compression should aim close to the maximum file size while remaining safely below it and preserving the required dimensions and format.
Every edit triggers revalidation	Any crop, background or enhancement change can affect dimensions and file size. The final file must be reprocessed and checked again.
Official technical constraints are data	Dimensions, size, format, background and naming rules should be stored as structured fields, not buried in code or prose.
Graceful uncertainty	When an exam requirement is absent or ambiguous, the system uses a documented standard process and clearly labels what was inferred.
No false guarantee	The platform prepares the photo according to stored requirements; acceptance remains with the official portal or authority.
Fast by default	The standard flow should require no manual editing. Manual controls are a later enhancement, not a dependency for the core experience.

Confirmed interpretation of 75-80% coverage

The crop engine should not reject a good image because a calculated percentage is marginally outside a narrow band. It should use face and head geometry to create a naturally tight, exam-appropriate crop that ordinarily lands around the expected coverage.

4. End-to-end user experience

The one-minute user journey



4.1 Step-by-step flow

Stage	Required experience
Discover or open the service	The landing page explains the promise in simple language and offers a prominent exam search.
Select the examination	The candidate chooses the exact exam and, where needed, the year or application cycle.
Review a concise requirement summary	Show only essential information: background, format, dimensions or range, file-size limit and any special instruction.
Upload a source photo	Accept common source formats. Provide a short checklist before upload: one person, face forward, adequate light, complete head visible and no severe blur.
Run pre-check	Verify that one face is present and that the image has enough information for a clean result.
Process automatically	Apply background clearance, intelligent crop, restrained correction, resize, compression, naming and final validation.
Show preview	Present the final image, not a confusing editing workspace. Pair it with a compact validation summary.
Download	Deliver the final file with the correct extension and filename.
Recover from a poor result	In the MVP, allow re-upload or automatic reprocessing. A manual crop/background editor can be introduced later as a pro feature.

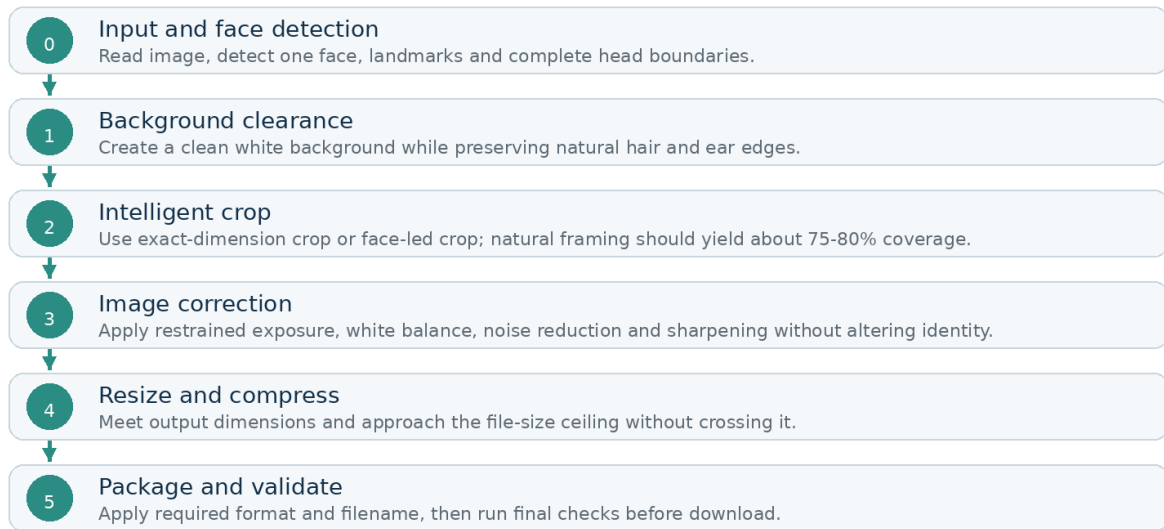
4.2 Preview and validation summary

The preview should communicate confidence without overwhelming the candidate. A useful summary can show the selected exam, output dimensions, final file size against the limit, format, background status, crop status and filename. Avoid exposing low-level model scores unless they help support or diagnose a problem.

Example check	Example display
Dimensions	300 x 400 px - matched
File size	47.8 KB of 50 KB maximum
Format	JPEG - matched
Background	White - processed
Framing	Head fully visible; natural tight crop
Filename	photograph.jpg

5. Image-processing engine and crop logic

Processing engine: ordered pipeline with revalidation



5.1 Stage 0 - Input reading and face detection

The engine begins by normalizing image orientation and reading metadata safely. It detects the face, facial landmarks and the larger head region. Face detection alone is not enough because cropping must also protect hair, ears, chin and beard boundaries.

- Detect exactly one primary face and estimate pose, eye line and face center.
- Estimate the full head boundary, including hair volume beyond the facial oval.
- Check whether the source frame has already clipped the top of the head, ears or chin.
- Estimate blur, resolution, overexposure, underexposure and major occlusion.
- Classify the input as processable, processable with caution, or unsuitable.

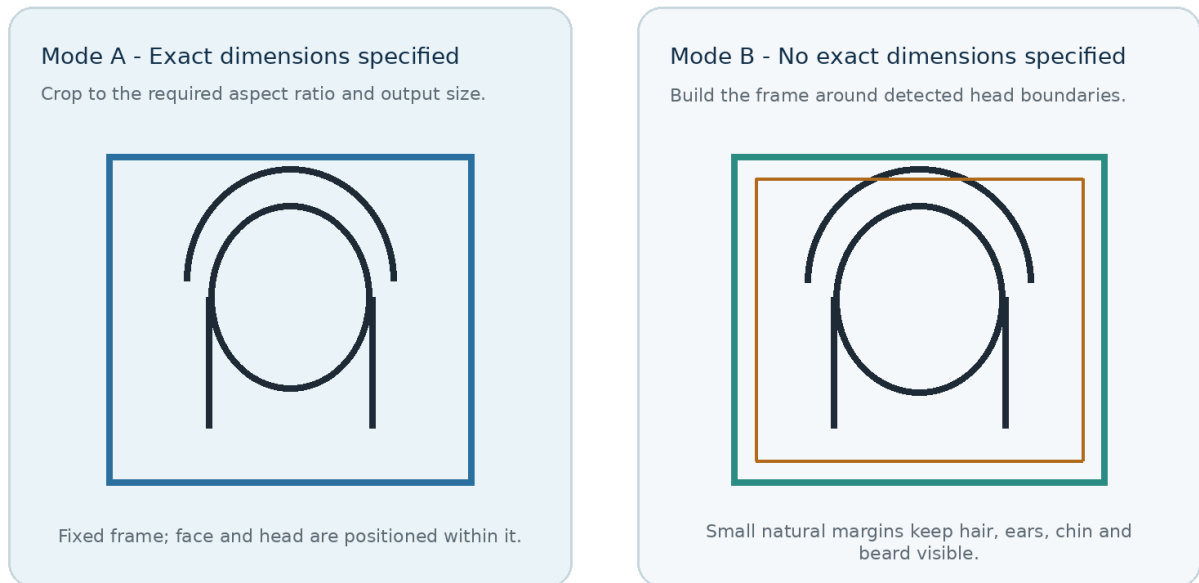
5.2 Stage 1 - Background clearance

Most supported exam photos are expected to use a plain light or white background. The engine should separate the person from the background, rebuild the background as the required color, and preserve edge detail.

Requirement	Expected behavior
Hair edges	Preserve fine strands where possible; avoid a painted or helmet-like edge
Ear and jaw edges	Keep natural contours without hard cut-outs
Halo control	Remove light or dark outlines caused by poor masking
Shadow handling	Reduce background shadows without flattening the face
Clothing boundary	Preserve shoulders and collar cleanly
Background color	Use the exam record; white is the platform default where the rule is unspecified

5.3 Stage 2 - Intelligent cropping: the most important processing decision

Two crop modes, one composition objective



Crop Mode A - Exact dimensions are specified

The engine creates the required aspect-ratio frame, positions the head within that frame, preserves complete head boundaries and then resizes to the exact output dimensions. The face cannot simply be stretched to fit. Repositioning and crop-window selection must happen before resizing.

Crop Mode B - Exact dimensions are not specified

The engine builds a crop around the detected head. It leaves a thin but natural margin on the top, sides and bottom so that the hair, ears, chin and final beard line remain visible. If the official rule provides a pixel range, the system chooses an output within that range after determining the crop.

Universal crop objective

In both crop modes, correct composition should ordinarily produce approximately 75-80% face coverage. The engine should optimize for natural, complete and centered framing rather than enforce a brittle one-number cutoff.

5.4 Defining coverage for implementation

The phrase "face coverage" can be interpreted differently by different teams. Before model tuning, the engineering and product teams must agree on a measurable internal definition. The most practical approach is to track several ratios rather than one ambiguous number: face bounding-box height to image height, head bounding-box height to image height, face width to image width, and top/bottom margin ratios. The user-facing product does not need to expose these measurements.

Internal measurement	Purpose
Head-height ratio	Controls overall tightness while accounting for hair
Face-height ratio	Approximates the commonly stated face-coverage requirement
Side-margin ratio	Prevents ears or hair from touching the frame
Top-margin ratio	Ensures the top of the hair is visible
Bottom-margin ratio	Keeps chin and beard line visible without excessive torso

5.5 Stage 3 - Restrained image correction

Corrections should improve legibility and consistency, not appearance in a cosmetic sense. A successful result still looks like the original person under neutral lighting.

Allowed	Not allowed
Exposure and brightness correction	Changing face shape or slimming the jaw
White balance and neutral color correction	Skin whitening or major skin-tone shifts
Mild noise reduction	Removing permanent marks or distinctive features
Mild sharpening after resize	Inventing eyes, hair or facial detail
Local shadow control when needed	Changing expression, gaze, pose or identity

5.6 Stage 4 - Resize and quality-aware compression

Compression is not a single fixed quality setting. The engine should search for the highest practical encoded quality that produces a file below the maximum limit. It should include a safety margin because different systems may report file size differently or add metadata.

15. Resize to the required output dimensions or the selected dimensions within the allowed range.
16. Encode at a high initial quality.
17. Measure the actual byte size of the final file.
18. If the file is too large, lower quality gradually or optimize encoding parameters.
19. If the file is far below the ceiling, raise quality where useful.
20. Strip unnecessary metadata while preserving correct orientation.
21. Stop at the best visual result below the configured safe ceiling.
22. Run a final decode test to confirm the file is valid and readable.

5.7 Stage 5 - Naming, packaging and download

The output service applies the required extension and filename pattern. When the exam does not specify a filename, the platform uses a safe standard such as candidate_photo.jpg. Filenames must avoid unsupported characters and accidental double extensions.

5.8 Reprocessing after user changes

Manual editing is deferred from the first release, but the processing contract should support it later. After any crop move, background rework or other image change, the system must repeat resize, compression, filename packaging and final validation. An edited preview must never be downloaded without this final pass.

6. Exam-guideline data foundation

6.1 Why the rules database is a product, not a support file

The engine can only be exam-specific if the requirements are collected, structured, versioned and maintained. A spreadsheet copied once from websites will quickly become unreliable. The data layer needs ownership, source evidence, status and review dates.

6.2 Core exam record

Data group	Fields and purpose
Identity	Exam name, conducting body, exam code, year/cycle and aliases
Source evidence	Official source URL or document, document title, page/section and captured text
Dimensions	Exact width/height, allowed width/height ranges, unit and aspect ratio
File size	Minimum KB if any, maximum KB, target safety margin
File format	Allowed formats and preferred export format
Background	Required color and any wording about plain/light background
Composition	Any explicit face coverage, head position, ear visibility or recent-photo instruction
Filename	Exact filename, pattern, case sensitivity and allowed characters
Exceptions	Spectacles, headwear, date/name imprint, black-and-white restrictions or portal-specific notes
Lifecycle	Status, verified date, effective cycle, superseded record and reviewer

6.3 Rule states

State	Meaning	Product behavior
Verified	Confirmed from an official, current source	Use automatically and display as verified
Verified with ambiguity	Official source exists but one field is unclear	Use explicit fields; mark inferred fields internally
Provisional	Collected from a secondary or older source	Do not silently present as current; prioritize review
Expired	Belongs to a closed or superseded cycle	Retain for history; do not use for new processing
Draft	Research in progress	Not selectable by public users

6.4 Fallback standard when a field is unspecified

A fallback must be deterministic and documented. It should not invent an official rule. It should only decide how the platform behaves when the official source does not specify a field.

Unspecified field	Recommended fallback behavior
Exact dimensions	Use a portrait frame derived from natural head-led cropping, then export a practical standard size
Background	Use clean white
Filename	Use candidate_photo with the chosen extension
Preferred format	Use JPEG unless transparency or another explicit requirement applies

Unspecified field	Recommended fallback behavior
Minimum file size	Optimize visual quality below the maximum; do not pad the file artificially
Coverage wording	Apply the platform's natural tight-crop objective, ordinarily around 75-80%

Data principle

Fallback values are platform defaults, not claims about an exam authority. The admin system must store whether each value is official, inferred or defaulted.

6.5 Research and verification workflow

23. Create a prioritized exam inventory based on application volume and launch relevance.
24. Collect the latest official notification, application instructions or upload page for each cycle.
25. Capture the exact wording and source location for every photo rule.
26. Enter normalized values into the structured schema while retaining the original text.
27. Have a second reviewer verify high-impact fields such as maximum size and dimensions.
28. Publish the record only after required fields and evidence are complete.
29. Re-check the record when a new application cycle opens or the official portal changes.
30. Preserve older versions for audit and reproduction of past outputs.

6.6 Admin console requirements

- Search, create, edit, duplicate and version exam records.
- Attach source links, document references and captured rule text.
- Preview how a rule record configures the image pipeline.
- Validate contradictory fields before publishing.
- Show records due for re-verification.
- Allow rollback to a previous verified version.
- Keep a change log with editor, timestamp and reason.

7. Validation, quality and failure handling

7.1 Three levels of validation

Level	Question answered	Examples
Input suitability	Can this photo produce a credible result?	One face, sufficient resolution, complete head, manageable pose and blur
Processing quality	Did each transformation work visually?	Clean mask, natural edges, centered crop, acceptable lighting
Output compliance	Does the file meet configured technical rules?	Dimensions, bytes, format, filename and background

7.2 Input conditions that should trigger guidance

Condition	Recommended response
No face detected	Ask for a clearer, front-facing photo
Multiple faces	Ask for a photo containing only the candidate
Head already clipped	Explain that missing hair, ear or chin detail cannot be reconstructed reliably
Severe blur or very low resolution	Ask for a sharper or larger image
Extreme side pose or tilted head	Ask for a front-facing, naturally aligned image
Strong obstruction	Ask the user to remove objects covering important facial regions where appropriate
Very uneven lighting	Attempt correction when safe; otherwise request a better-lit image
Background segmentation uncertainty	Retry with alternate masking settings; if still poor, ask for a cleaner source photo

7.3 Do not over-reject

The input checker should protect output quality without becoming a barrier. Mild lighting issues, ordinary backgrounds, small pose differences and normal variations in face shape should be handled by the engine. Rejection is appropriate only when essential information is missing or the result would be visibly unreliable.

7.4 Quality scoring

Internally, the system may calculate confidence scores for detection, segmentation, crop safety and final clarity. These should control retries, fallback algorithms and error routing. The public interface should translate them into plain guidance rather than raw percentages.

7.5 Final validation contract

Check	Pass condition
File integrity	Image decodes successfully and is not corrupted
Format	Matches an allowed exam format
Dimensions	Exact values or within configured range
File size	Within permitted range and below maximum safety ceiling
Filename	Matches configured pattern or fallback standard
Background	Matches configured color with no major residual clutter
Framing	Complete head boundaries, centered face and natural tight crop
Identity fidelity	No geometry or appearance changes that alter recognition

8. Product scope, MVP and roadmap

8.1 MVP objective

Prove that a candidate can select a supported exam, upload a suitable photo, and receive a high-quality exam-configured output without manually editing the image.

8.2 MVP - included

- Exam search and selection for a curated launch set.
- Versioned exam-rule records managed through an internal admin interface.
- Upload of common image formats with orientation handling.
- One-face and source-suitability pre-check.
- White or configured background clearance.
- The two crop modes defined in this document.
- Restrained exposure and clarity correction.
- Exact resize or range-based output sizing.
- Quality-aware compression close to the maximum file-size ceiling.
- Required output format and filename handling.
- Preview, concise validation summary and download.
- Re-upload and automatic reprocess when the source is unsuitable or the result is poor.
- Short-lived storage and automatic deletion.

8.3 MVP - explicitly excluded

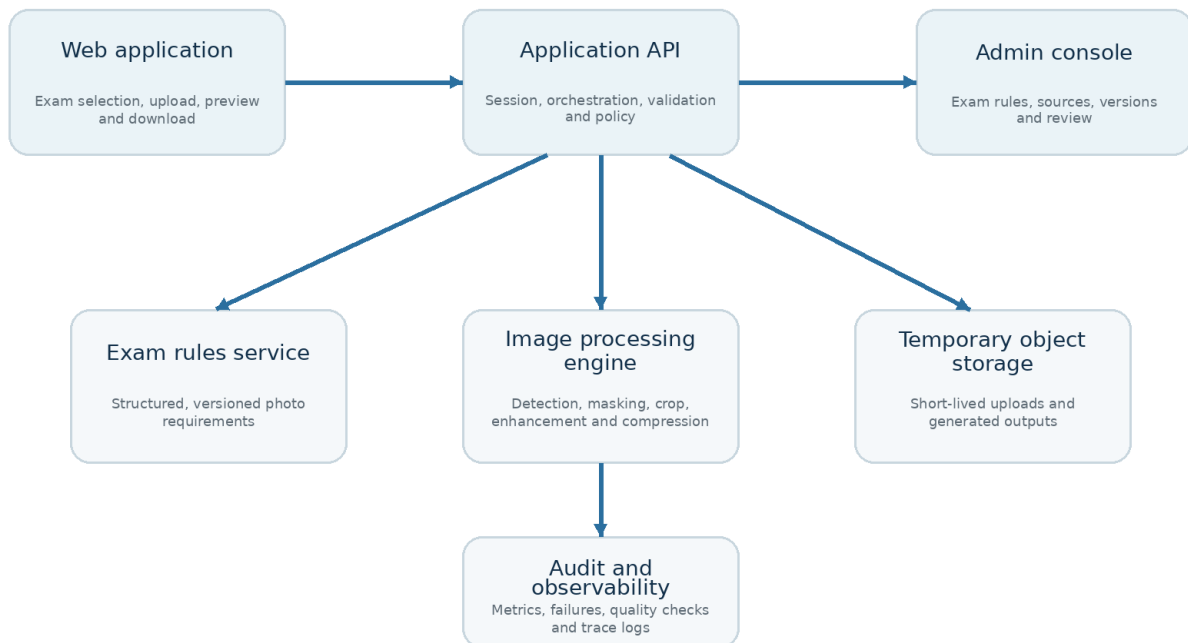
- Payment gateway and paid download.
- Full manual crop editor.
- Background brush, erase/restore tools or local retouching.
- Cosmetic editing or beautification.
- Bulk processing for operators.
- Native mobile apps.
- Guaranteed acceptance claims.
- Coverage of every Indian examination on launch day.

8.4 Phased roadmap

Phase	Goal	Key deliverables
Phase 0 - Foundation and prototype	Validate crop, masking and compression quality	Sample rule schema; benchmark set; processing prototype; crop definition
Phase 1 - Focused MVP	Deliver the one-click journey for a curated exam set	Public web flow; processing API; rules admin; preview and download
Phase 2 - Coverage and reliability	Expand exam catalog and harden operations	Research workflow; versioning; monitoring; retry paths; broader test matrix
Phase 3 - Controlled editing	Give users precision controls when automation is not enough	Manual crop adjustment; background rework; automatic revalidation
Phase 4 - Commercial layer	Monetize enhanced value	Payments; pro edits; bundles; operator tools; account history

9. Recommended system architecture

Recommended logical architecture



9.1 Logical components

Component	Responsibility
Web application	Exam search, upload, progress, preview, validation summary and download
Application API	Sessions, authorization where needed, rule retrieval, processing orchestration and output policy
Exam rules service	Versioned structured requirements and source evidence
Image processing engine	Detection, segmentation, crop, enhancement, resize, compression and validation
Temporary object storage	Short-lived originals, intermediate images and final outputs
Admin console	Research entry, review, publishing, versioning and audit
Observability layer	Timing, errors, quality flags, retry outcomes and system health

9.2 Processing deployment choices

The image engine may run synchronously for small, fast jobs, but the architecture should be ready for asynchronous processing if background segmentation or retries become slow. The public experience can still feel instant by showing clear progress and polling or streaming the result.

9.3 Recommended request flow

31. The client requests the selected exam's current public rule summary.
32. The server creates a short-lived processing session and upload authorization.
33. The source image is uploaded directly to temporary storage or through the API.
34. The orchestration service invokes the processing engine with an immutable rule snapshot.
35. The engine writes a final output and machine-readable validation report.
36. The API exposes an authenticated preview and short-lived download link.
37. The system schedules deletion according to the retention policy.

38. Logs retain technical metadata and failure reasons, not the image itself after deletion.

9.4 Rule snapshotting

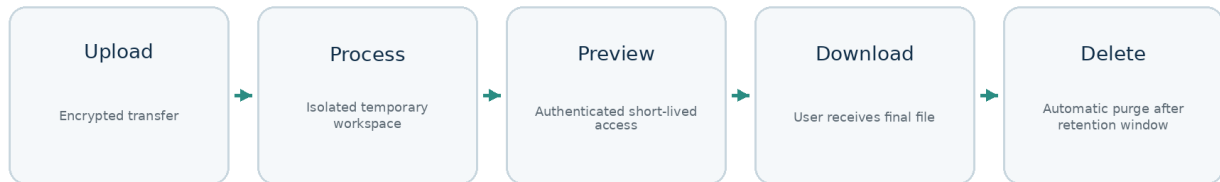
Every processing job should store the exact version of the exam rule used. If an admin changes a rule later, the platform must still be able to explain how an earlier file was produced. This is essential for debugging and trust.

9.5 Technology selection criteria

Area	Selection criteria
Face and landmark detection	Reliable on Indian users, varied lighting, hairstyles, beards and mobile images
Segmentation	High-quality hair and ear edges; acceptable CPU/GPU cost
Image operations	Deterministic resize, color conversion, metadata stripping and compression control
Database	Strong relational structure, JSON support where useful and clean version history
Storage	Private objects, signed URLs, lifecycle deletion and regional compliance options
Hosting	Low latency for Indian users, observability, secure secrets and predictable cost

10. Privacy, security and responsible image processing

Privacy-first image lifecycle



10.1 Why privacy is foundational

A passport-style photograph is personal data and may be used in high-stakes identity workflows. The platform should minimize collection, minimize retention and clearly explain what happens to every upload.

10.2 Minimum privacy requirements

- Use encrypted transport for uploads and downloads.
- Keep image objects private; never expose permanent public URLs.
- Use short-lived signed access for preview and download.
- Delete originals, intermediates and outputs automatically after a clearly stated period.
- Do not use uploaded photos for model training without explicit, separate consent.
- Do not retain facial embeddings unless a specific, justified feature requires them.
- Keep operational logs free of raw images and unnecessary personal identifiers.
- Provide a direct delete action where technically feasible.
- Limit staff access and record administrative access to stored images.
- Publish a clear privacy notice before public launch.

10.3 Responsible enhancement policy

The processing engine must preserve identity. Any future AI-based enhancement model should be tested for hallucination, skin-tone shifts, facial-feature changes and uneven performance across genders, complexions, hairstyles, facial hair, ages and assistive or religious appearance.

10.4 Security baseline

Control	Purpose
Content-type and file-signature validation	Prevent disguised or malicious uploads
Image decode sandboxing	Reduce risk from malformed image files
Maximum input limits	Protect compute and storage from abuse
Rate limiting	Control automated misuse and cost spikes
Virus/malware scanning where appropriate	Protect systems and downstream users
Secret management	Keep storage and model credentials out of client code
Audit trails	Support incident investigation and rule-change accountability

11. UX structure and screen requirements

11.1 UX character

The product should feel official enough to inspire trust, but not bureaucratic. The interaction should be calm, guided and extremely clear. The interface should not resemble a complex photo editor during the standard flow.

11.2 Minimum public screens

Screen	Primary purpose	Essential elements
Landing / exam search	Explain value and start the flow	Exam search, supported-exam examples, simple three-step explanation, privacy reassurance
Exam requirement summary	Confirm the selected context	Exam/cycle, concise rules, change-exam action, upload CTA
Upload	Get a processable source image	Drop zone, camera/file option, source-photo checklist, privacy note
Processing	Maintain confidence during work	Progress stages, no fake percentages, cancel/retry handling
Result	Confirm and deliver	Large preview, validation summary, download, re-upload or reprocess
Error guidance	Help the user recover	Plain problem statement, visual tips, exact next action

11.3 Source-photo guidance

- Only one person in the photo.
- Face the camera with a natural, neutral expression.
- Keep the entire head, hair, ears and chin inside the source image.
- Use even light and avoid strong shadows across the face.
- Avoid severe blur, heavy filters and screenshots of very small images.
- Use a recent image that still represents the candidate accurately.

11.4 Result-screen hierarchy

39. Final image preview.
40. Download button with filename and file size.
41. Compact technical checks.
42. Selected exam and cycle.
43. Re-upload or automatic reprocess option.
44. Clear disclaimer that final portal acceptance is controlled by the examination authority.

11.5 Accessibility and mobile requirements

- The complete core journey must work on a small mobile screen.
- All status messages must be understandable without color alone.
- Controls need clear labels, keyboard access and visible focus states.
- Upload guidance and errors should use simple language and supporting examples.
- Progress and result states should be announced to assistive technology.
- Preview images require meaningful alternative text without exposing unnecessary personal details.

12. Development plan, QA and success metrics

12.1 Workstreams

Workstream	First deliverables
Product and UX	Clickable flow, requirement-summary pattern, result-state design and error language
Exam data	Schema, admin workflow, first verified exam set and review checklist
Computer vision	Detection benchmark, segmentation comparison, crop algorithm and quality thresholds
Image engineering	Resize/compression search, format conversion, metadata handling and validation
Backend platform	Job orchestration, storage lifecycle, signed access and rule snapshotting
Frontend platform	Upload, progress, preview, download and mobile behavior
QA and operations	Golden image set, portal tests, monitoring, support process and release gates
Legal and trust	Privacy policy, terms, disclaimer and consent language

12.2 Recommended development sequence

45. Freeze the rule schema and the internal definition of crop quality.
46. Build a benchmark dataset of consented or synthetic test images covering varied faces, hair, beards, lighting and backgrounds.
47. Prototype face/head detection and the two crop modes before building the full website.
48. Prototype background segmentation and compare edge quality on difficult hair and ear cases.
49. Build deterministic resize, compression and final validation as a standalone library.
50. Integrate the image pipeline behind a simple API using immutable rule snapshots.
51. Build the admin console and seed a small, verified exam catalog.
52. Build the public one-click flow and instrument every stage.
53. Run portal-level tests with representative rule combinations.
54. Launch gradually, inspect failure clusters and expand exam coverage only after reliability is stable.

12.3 Golden test set

The team should maintain a fixed test suite that runs after every meaningful engine change. It should include normal and difficult cases, not only ideal studio photos.

Test dimension	Required variety
Face and appearance	Different face shapes, skin tones, ages, genders, hairstyles, bald heads, beards and moustaches
Clothing and accessories	Light/dark clothing, collars, spectacles and permitted head coverings
Background	Plain, cluttered, bright, dark, patterned and shadowed
Lighting	Even, underexposed, overexposed, side-lit and mixed-color light
Capture	Mobile camera, scanned photo, screenshot, social-media export and rotated metadata
Framing	Loose portrait, tight portrait, off-center face and limited margins
Rule combinations	Exact size, size range, multiple formats, very small KB ceiling and filename pattern

12.4 Release acceptance criteria

- A supported exam record cannot be published without source evidence and required technical fields.
- The system consistently preserves complete head boundaries on the golden set where source information exists.
- Background edges do not show obvious hard cuts, halos or large missed regions at normal viewing size.
- Outputs meet configured dimensions, format and byte limits in automated tests.
- Compression never crosses the configured safe maximum.
- Identity-changing enhancement is absent in human review and regression tests.
- Files download with the correct extension and valid filename.
- Originals and generated files are deleted according to policy.
- Common failures produce actionable, non-technical guidance.
- The complete flow works on representative Android phones and desktop browsers.

12.5 Product success metrics

Metric	What it reveals
Processing completion rate	Whether users can reach a final output
First-attempt download rate	Whether the automatic result is satisfactory
Re-upload or retry rate	Where source guidance or engine quality is weak
Median and 95th percentile processing time	Whether the experience feels immediate
Technical pass rate by exam rule	Whether dimensions, bytes, format and naming are reliable
Background/crop complaint rate	Whether the two highest-risk visual stages are working
Support contacts per 1,000 jobs	Whether the experience is understandable
Exam-rule correction frequency	Whether the research process and versioning are reliable
Deletion-policy success rate	Whether privacy operations are functioning

13. Risks, open decisions and decision ledger

13.1 Principal risks and mitigations

Risk	Impact	Mitigation
Outdated exam rules	Technically perfect image may follow an old requirement	Version by cycle, retain source evidence, re-verify at application opening
Poor hair-edge segmentation	Output looks edited or unofficial	Benchmark multiple models; use edge refinement and automatic retry
Ambiguous 75-80% wording	Teams implement inconsistent crop calculations	Define internal ratios and approve a visual crop standard
Tiny file-size ceilings	Quality degrades visibly	Use quality search, optimized encoding and realistic warnings
Overprocessing changes identity	Trust and eligibility risk	Strict enhancement policy, regression tests and human review
User uploads unsuitable source	Automation cannot recover missing detail	Friendly pre-check and clear re-upload guidance
Slow processing cost	Poor UX or unsustainable unit economics	Profile stages, cache models, choose CPU/GPU path and queue when needed
False acceptance expectation	User trust and legal risk	Careful claims, visible rule source/date and authority disclaimer
Sensitive image retention	Privacy and security harm	Short retention, private storage, lifecycle deletion and minimal logs

13.2 Open product decisions to resolve during Phase 0

Decision	Why it matters	Recommended next action
Exact internal crop measurements	Needed for consistent engineering and QA	Label a benchmark set and compare ratio definitions against human-approved crops
Launch exam list	Defines research and QA scope	Prioritize a small set by relevance and rule diversity
Default dimensions when unspecified	Affects fallback consistency	Choose a portrait standard after testing quality and common portal behavior
File-size safety margin	Prevents boundary failures while preserving quality	Test portal behavior and byte reporting; configure per rule if needed
Image retention period	Balances user convenience and privacy	Choose the shortest period compatible with download and support
Processing latency target	Guides architecture and model choice	Measure prototype stages on realistic infrastructure
MVP reprocess behavior	Determines recovery without a manual editor	Define automatic retry options and re-upload UX
Authentication requirement	Affects friction and privacy	Prefer no account for basic MVP unless history or payment requires it

13.3 Foundational decision ledger

Status	Decision
CONFIRMED	The user selects an exam before uploading the photo.
CONFIRMED	The engine handles background, cropping, restrained enhancement, dimensions, file size, format and filename.
CONFIRMED	There are two crop modes: exact-dimension crop and face-led crop when exact dimensions are absent.
CONFIRMED	Correct natural cropping should ordinarily produce approximately 75-80% face coverage in all cases.
CONFIRMED	Hair, ears, chin and final beard line must remain visible with appropriate margins.
CONFIRMED	Compression should approach the maximum allowed size without crossing it.
CONFIRMED	Exam requirements must be researched and stored structurally.
CONFIRMED	Payments are not part of the first release.
CONFIRMED	Advanced manual crop/background editing is a later pro capability.
RECOMMENDED	The MVP should still allow re-upload and automatic reprocessing.
RECOMMENDED	Every job should retain the exact rule-version reference but not retain the image beyond the deletion window.
RECOMMENDED	The product should avoid guaranteed-acceptance claims.

Appendix A - Suggested exam-rule schema

The following is a conceptual schema. Field names may change, but the information categories should remain. Values that come from an official source should be distinguished from platform defaults or internal inference.

```

exam_id: string
exam_name: string
conducting_body: string
application_cycle: string
status: draft | verified | provisional | expired

photo_rule:
  background_color: white | light | custom
  allowed_formats: [jpg, jpeg, png]
  preferred_format: jpg
  width_px_exact: integer | null
  height_px_exact: integer | null
  width_px_min: integer | null
  width_px_max: integer | null
  height_px_min: integer | null
  height_px_max: integer | null
  file_size_kb_min: number | null
  file_size_kb_max: number
  filename_pattern: string | null
  official_composition_text: string | null
  special_instructions: [string]

processing_policy:
  crop_mode: exact_dimensions | face_led
  target_composition: natural_tight_crop_75_80
  safety_margin_bytes: integer
  fallback_fields_used: [string]

source:
  source_type: official_page | notification | portal_instruction
  source_reference: string
  source_excerpt: string
  effective_from: date | null
  verified_at: datetime
  verified_by: user_id
  supersedes_rule_version: string | null

```

Appendix B - Processing job contract

Input	Required contents
Rule snapshot	Immutable exam-rule version and processing policy
Source image	Private object reference, detected format and orientation
Session policy	Retention window, retry allowance and output access scope

Output	Required contents
Final image	Private object reference, format, dimensions and bytes
Validation report	Pass/fail checks, crop metrics, quality flags and fallback use
Processing trace	Stage durations, algorithm versions, retries and safe error codes
Deletion schedule	Timestamp for source, intermediate and output purge

Appendix C - MVP acceptance checklist

- ☐ Exam selection retrieves the intended current verified rule.
- ☐ Upload supports rotation metadata and validates the actual image signature.
- ☐ Exactly one primary face is detected or the user receives clear guidance.
- ☐ Complete head boundaries are preserved whenever present in the source.
- ☐ Background is clean and required edge quality is met.
- ☐ Crop mode is selected correctly from the exam record.
- ☐ Natural tight framing is achieved, ordinarily around 75-80% face coverage.
- ☐ Output dimensions match exact values or configured ranges.
- ☐ Output file size is below the safe maximum and quality is not unnecessarily reduced.
- ☐ Output format and filename are correct.
- ☐ Final file decodes successfully after packaging.
- ☐ Preview and validation summary show consistent values.
- ☐ Re-upload and automatic retry paths work.
- ☐ Private links expire and storage lifecycle deletion succeeds.
- ☐ No facial geometry or identity-changing enhancement is applied.
- ☐ Mobile and desktop flows pass accessibility and usability checks.
- ☐ Claims and disclaimers accurately describe the service.

Appendix D - Glossary

Term	Meaning in this product
Exam rule	A structured requirement used to configure output processing
Rule snapshot	The exact immutable rule version attached to one processing job
Face detection	Locating the facial region and landmarks
Head detection	Estimating the full head including hair and ear boundaries
Segmentation	Separating the person from the original background
Exact-dimension crop	Crop constrained by a required output aspect ratio and size
Face-led crop	Crop built around detected head geometry when exact dimensions are absent
Natural tight crop	A centered crop with small margins and complete head visibility, ordinarily yielding about 75-80% face coverage
Safe ceiling	A configured file size below the official maximum to prevent accidental overflow
Validation report	Machine-readable result of input, processing and output checks
Fallback	A documented platform behavior used only when an official field is unspecified

Final foundation statement

The product succeeds when a candidate can make one clear choice, provide one suitable photo and receive one trustworthy, exam-configured file - without learning image-editing terminology or moving between multiple tools.